Bank Term Deposit Subscription Prediction: A Comparative Study of Five Machine Learning Models on Imbalanced Telemarketing Data

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Abstract. We study whether a bank can identify in advance which customers will subscribe to a term deposit, using 41,188 records from the UCI Bank Marketing Dataset [12,13], where only 11.27% of contacts resulted in a subscription. Five classifiers: Logistic Regression, Decision Tree, Random Forest [2], XGBoost [4], and LightGBM [9] followed an identical pipeline: stratified train/validation/test split, feature selection by mutual information, scaling, and SMOTE [3] + Tomek Links [16] fitted only within the training portion of each cross-validation fold. The final decision threshold is selected on the separate validation set by maximizing F1 subject to a Recall check of at least 0.50, before a single evaluation on the held-out test set.

We report two parallel scenarios. Scenario A retains call duration as a post-contact reference. Under five-fold cross-validation, LightGBM achieves the highest PR-AUC (0.6272), while XGBoost achieves the highest F1-score (0.6243). Scenario B excludes call duration because it is unavailable before contact and therefore represents the deployable setting. In Scenario B, LightGBM achieves the highest cross-validation PR-AUC (0.4239), F1-score (0.4746), and ROC-AUC (0.7824), so it is selected for subsequent tuning. After tuning and validation-based threshold selection, the final LightGBM model uses a threshold of 0.617 and achieves a held-out test PR-AUC of 0.4658, F1-score of 0.5071, Recall of 0.5399, Precision of 0.4781, and ROC-AUC of 0.8049. Ranking the same test customers with this deployable model shows that calling the top 20% captures 607 of 928 subscribers (65.41%) while avoiding approximately 80% of calls.

Keywords: Bank telemarketing · Term deposit subscription · LightGBM · Class imbalance · SMOTE · Customer ranking

1 Introduction

The UCI Bank Marketing dataset records telephone campaigns promoting term deposits to 41,188 customers between May 2008 and November 2010 [12,13]. Only 11.27% of contacts resulted in a subscription, making indiscriminate calling inefficient. A predictive ranking could help prioritize customers with higher estimated subscription probabilities.

The task presents two challenges. First, the positive class is rare (approximately 1:8), so Accuracy alone does not adequately describe minority-class performance. Second, call duration is strongly predictive but is known only after a call ends; it is therefore unavailable when deciding whom to contact.

Previous studies have examined conventional classifiers, tree ensembles, resampling, and cost-sensitive learning [1,6,7,8,10,17,19]. Their results are difficult to compare directly because data splits, feature sets, imbalance treatments, and the use of duration vary. This study compares five classifiers under one leakage-aware protocol, separates post-contact and pre-contact settings, and evaluates the operational value of ranking customers using only pre-contact information.

2 Related Work

Prior comparisons on bank-marketing data report mixed rankings: Du [7] found gradient boosting competitive under a leakage-aware protocol, Borugadda et al. [1] reported Logistic Regression as strongest, while studies using Kaggle variants often favored Random Forest [10,19]. These differences underline the sensitivity of results to data splits and evaluation protocols.

Recent studies address imbalance through feature selection, resampling, and cost-sensitive learning. Ghatasheh et al. [8] combined genetic feature selection with XGBoost; Chen and Luo [5] tuned oversampling for LightGBM; Dong et al. [6] combined partial SMOTE with cost-sensitive learning; and Wu [18] applied correlation-based screening before SMOTE. Wang et al. [17] further studied ensemble learning and explainable AI. In contrast, our contribution is a controlled five-model comparison with fold-local preprocessing, explicit pre-/post-contact scenarios, and ranked-calling evaluation.

3 Data and Research Methods

Figure 1 summarizes the leakage-aware experimental pipeline, including the stratified split, the two duration scenarios, fold-local preprocessing and resampling, cross-validation-based model selection, tuning, validation-based threshold selection, final test evaluation, and ranked-calling analysis. The subsections below describe each stage in more detail.

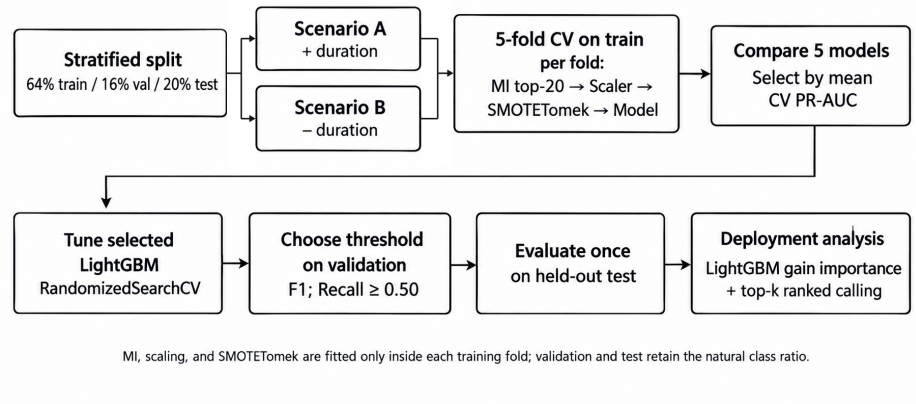


Fig. 1. Overview of the experimental pipeline. The data are split into 64% training, 16% validation, and 20% test subsets. Scenario A retains call duration, whereas Scenario B excludes it. Within each cross-validation fold, mutual-information feature selection, scaling, and SMOTETomek are fitted only on the fold's training portion. Scenario-B model selection uses mean CV PR-AUC, followed by tuning, validation-based threshold selection, one held-out test evaluation, feature interpretation, and top-k ranked calling.

3.1 Description of the dataset

This study uses the Bank Marketing dataset with economic and social context attributes, released on the UCI Machine Learning Repository [13] and compiled from a Portuguese bank telemarketing campaign between May 2008 and November 2010 [12]. The version used here contains 41,188 records with 20 input features and one binary target variable y indicating whether the client subscribed to a term deposit; the positive class covers about 11.27% of the records. There are no missing values, and missing information is coded as the category "unknown" and kept as a separate level. The features fall into four groups: customer information (age, job, marital status, education, default, housing, loan), current contact (channel, month, day of week, duration, campaign), past campaign history (pdays, previous, outcome), and macroeconomic indicators (emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed).

3.2 Data Preprocessing

Validation showed no empty cells; entries marked "unknown" were retained as a distinct category. Among the macroeconomic indicators, `emp.var.rate` and `nr.employed` show strong correlation with `euribor3m` ($r \geq 0.91$), while `cons.price.idx` ($r \approx 0.69$) and `cons.conf.idx` ($r \approx 0.28$) are only moderately and weakly correlated, respectively. We retain `euribor3m` as the primary macroeconomic indicator. `emp.var.rate` and `nr.employed` are removed because of their strong redundancy with it; `cons.price.idx` and `cons.conf.idx` are also excluded as part of a parsimonious macroeconomic feature-reduction decision rather than because of strong pairwise correlation.

The feature `duration` records total call length in seconds. As stated in the dataset documentation, this value is only known after a call concludes; therefore it cannot be used to prioritize customers before contact is made [12,13]. To separate predictive performance from practical deployability, the experiments use two parallel settings. Scenario A retains `duration` as a post-contact reference and provides an upper-bound comparison with studies that use this feature. Scenario B removes `duration` entirely, so all predictors are available before the call and the resulting model can be used for customer ranking. Performance is reported for both settings, while the ranked-calling analysis uses Scenario B only.

3.3 Exploratory data analysis

Before modeling, the data were examined at three levels. The label distribution showed that only 11.27% of the records belong to the positive class, which ruled out plain Accuracy and made F1-score and PR-AUC the primary metrics. Cross-tabulations by category revealed that students and retired customers subscribe at much higher rates, that calls placed at the turn of the year work better, and that longer conversations end in a subscription more often. Correlation analysis showed the economic indicators moving almost together and repeat contacts within one campaign lowering the acceptance rate, which informed the removal of redundant indicators. Bar charts, box plots, and distribution plots confirmed that no anomalous values or serious outliers required removal.

3.4 Feature engineering and selection

Age was grouped into four bands (under 25, 25–44, 45–64, and 65 or older), and month was compressed into four quarters. Two binary features, `has_prev_success` and `has_prev_failure`, were derived from the previous campaign outcome. The special value `pdays=999` indicates that a customer had not previously been contacted in a previous campaign. To represent this distinction explicitly, we created a binary feature, `previously_contacted`, which equals 1 when `pdays` differs from `999` and 0 otherwise. We then replaced the special value `999` in `pdays` with `-1`, allowing the model to distinguish customers without a previous contact from those with a recorded number of days since their last contact. This transformation preserves the information about previous contact status while providing an explicit indicator for the model. After the train/validation/test split, education and the remaining categorical variables were one-hot encoded; validation and test matrices were then reindexed to the training-set feature columns. The target was mapped to 0 and 1. During model fitting, the twenty most informative features were selected by mutual information within the training portion of each cross-validation fold.

3.5 Data split and cross-validation

The complete dataset was stratified into three subsets in two stages: first, 20% was reserved as the held-out test set; the remaining 80% was further split 80:20 into training and validation sets. This yields approximately 64% training, 16% validation, and 20% test samples, with stratification preserving the original class ratio throughout. Within each model pipeline, mutual-information feature selection and StandardScaler were fitted only on the training portion of each cross-validation fold. Scaling was performed before SMOTETomek because SMOTE relies on distances between samples. Stratified five-fold cross-validation on the training set supported model comparison; the model with the highest mean CV PR-AUC in Scenario B was passed to hyperparameter tuning. The separate validation set guided threshold selection, and the held-out test set was reserved for final evaluation.

3.6 Class-imbalance handling

Class imbalance was handled using SMOTETomek from imbalanced-learn [11] inside the model pipeline. SMOTE [3] generated synthetic minority samples with `k_neighbors=5` and `sampling_strategy=0.7`, after which Tomek Links [16] removed ambiguous boundary pairs. With this sampling strategy, the minority-to-majority ratio is targeted at 0.7 before Tomek cleaning (approximately 41% minority samples in the resampled set before Tomek removal), rather than a 50:50 balance. Resampling is fitted only on the training portion of each cross-validation fold and is never applied to the validation or test sets. For the tuned LightGBM, the search selected `sampling_strategy=0.8`, corresponding to approximately 44% minority samples before Tomek removal (see Section 3.7).

Table 1. Model hyperparameter configurations

Model	Configuration
Logistic Regression	`max_iter`=3000, solver=`'liblinear'`, default $C = 1.0$ (linear baseline)
Decision Tree	`max_depth`=8, `min_samples_leaf`=10, `criterion='gini'`, `class_weight=None`
Random Forest	`n_estimators`=300, `max_depth`=12, `min_samples_split`=10, `class_weight=None` [2]
XGBoost	`n_estimators`=300, `max_depth`=6, `learning_rate`=0.05, `subsample`=0.8, `colsample_bytree`=0.8, `reg_lambda`=2.0, `scale_pos_weight`=1 [4]
LightGBM	`n_estimators`=300, `max_depth`=6, `learning_rate`=0.05, `num_leaves`=16, `class_weight=None` [9]
LightGBM (tuned)	`n_estimators`=400, `max_depth`=5, `learning_rate`=0.01, `num_leaves`=15, `subsample`=0.7, `colsample_bytree`=0.8, `reg_lambda`=1.0, `min_child_samples`=30, SMOTE `sampling_strategy`=0.8

No additional class weighting is used in the main five-model comparison so that all classifiers share the same imbalance-handling mechanism. Logistic Regression, Decision Tree, Random Forest, and LightGBM use no class weighting; XGBoost uses `scale_pos_weight=1`, which is equivalent to no extra positive-class weighting. Cost-sensitive weighting is therefore a future extension rather than part of the reported main experiment.

3.7 Machine learning models and threshold selection strategy

Five classifiers were trained and compared under identical experimental settings. After the five-model comparison, only the Scenario-B model selected by mean CV PR–AUC was tuned. `RandomizedSearchCV` evaluated 12 candidate configurations with four-fold cross-validation and average precision as the objective; four folds were used at this stage to reduce tuning cost while keeping model comparison itself at five folds. For LightGBM, the search space was: SMOTE `sampling_strategy` $\in \{0.6, 0.7, 0.8\}$; `n_estimators` $\in \{400, 600, 800, 1000\}$; `max_depth` $\in \{4, 5, 6, 8\}$; `learning_rate` $\in \{0.01, 0.02, 0.03, 0.05\}$; `num_leaves` $\in \{15, 31, 50\}$; `subsample` and `colsample_bytree` $\in \{0.7, 0.8, 0.9\}$; `reg_lambda` $\in \{1.0, 2.0, 3.0\}$; and `min_child_samples` $\in \{20, 30, 50\}$. The selected LightGBM configuration is reported in Table 1. Decision thresholds were then evaluated on the validation set. The unconstrained F1 optimum and the highest-F1 threshold among points with Recall at least 0.50 were identical after rounding, giving a final threshold of 0.617. This threshold was applied unchanged to the test set.

3.8 Experimental setup and evaluation metrics

All experiments ran on Google Colab with Python 3 and scikit-learn [14], using `SEED=42`. Performance is reported using Precision, Recall, F1-score, Balanced Accuracy, ROC-AUC, and PR-AUC. Precision and Recall are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{1}$$

Table 2. Five-fold cross-validation results of the five models on the training set, Scenario A (duration retained). SMOTE+Tomek is fitted inside each fold; validation folds keep the natural class ratio and are scored at the default 0.5 threshold. Column-wise best in bold. Time is the wall-clock time of the cross-validation run.

Model	Accuracy	PR-AUC	F1	ROC-AUC	Time (s)
Logistic Regression	0.8656	0.5684	0.5813	0.9274	80.2
Decision Tree	0.8671	0.5732	0.5874	0.9267	64.3
Random Forest	0.8726	0.6135	0.6007	0.9356	96.7
XGBoost	0.8888	0.6199	0.6243	0.9407	70.2
LightGBM	0.8853	0.6272	0.6202	0.9407	75.9

and their harmonic mean is

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2}$$

Balanced Accuracy averages sensitivity and specificity. ROC-AUC measures discrimination across thresholds, whereas PR-AUC summarizes the precision–recall trade-off and is the primary selection metric because only 11.27% of records are positive [15]. Models are selected by mean five-fold CV PR-AUC; thresholds are chosen on validation data before the single test evaluation.

4 Research Results

Results are reported for the post-contact reference (Scenario A) and the deployable pre-contact setting (Scenario B).

4.1 Model performance

Table 2 reports the five-fold cross-validation results for Scenario A. LightGBM achieves the highest mean PR-AUC at 0.6272, followed by XGBoost at 0.6199 and Random Forest at 0.6135. XGBoost records the highest F1-score (0.6243) and Accuracy (0.8888), while LightGBM follows closely with F1 = 0.6202 and Accuracy = 0.8853. Both boosting models reach ROC-AUC = 0.9407. Decision Tree and Logistic Regression have the lowest PR-AUC values, at 0.5732 and 0.5684, respectively.

Scenario A is the post-contact reference, so it is reported first and Scenario B (the deployable setting) follows. Table 3 reports the test-set evaluation when call duration is kept as a predictor. XGBoost has the best PR-AUC (0.6643, 5.9 times the base rate), only 0.0020 ahead of LightGBM (0.6623), and the best Accuracy (0.9130) and Precision (0.5893). LightGBM has the best F1 (0.6610 versus 0.6600) and Balanced Accuracy (0.8576), and Random Forest the highest Recall (0.8114) at the lowest Precision (0.5084). Logistic Regression (0.6044) and the Decision Tree (0.5991) are the weakest in PR-AUC. The F1-optimal

Table 3. Test-set performance of the five models in Scenario A (duration retained, post-contact reference), each at its F1-optimized decision threshold from the validation stage (column-wise best in bold). These results are an upper bound and are not deployable, because duration is known only after the call.

Model	Thresh.	Accuracy	Bal. Acc.	PR-AUC	F1	Recall	Prec.
XGBoost	0.4805	**0.9130**	0.8418	**0.6643**	0.6600	0.7500	**0.5893**
LightGBM	0.5288	0.9085	**0.8576**	0.6623	**0.6610**	0.7920	0.5671
Random Forest	0.5745	0.8904	0.8559	0.6506	0.6252	**0.8114**	0.5084
Logistic Regression	0.6224	0.8991	0.8387	0.6044	0.6295	0.7608	0.5369
Decision Tree	0.6944	0.9051	0.8139	0.5991	0.6230	0.6961	0.5637

Table 4. Five-fold cross-validation results of the five models on the training set, Scenario B (duration excluded). SMOTE+Tomek is fitted inside each fold; validation folds keep the natural class ratio and are scored at the default 0.5 threshold. Column-wise best in bold. Time is the wall-clock time of the cross-validation run.

Model	Accuracy	PR-AUC	F1	ROC-AUC	Time (s)
Logistic Regression	0.8080	0.4170	0.4194	0.7770	63.4
Decision Tree	0.8613	0.3838	0.4600	0.7650	63.7
Random Forest	0.8581	0.4218	0.4659	0.7807	89.5
XGBoost	0.8699	0.4181	0.4722	0.7784	70.4
LightGBM	0.8673	0.4239	0.4746	0.7824	68.1

thresholds range from 0.4805 to 0.6944 (Scenario B: 0.3997–0.5839, Table 5). Because duration is known only after a call ends [12,13], these figures are an upper bound and not a deployable result; the two scenarios are compared in Section 4.5.

Table 4 reports the five-fold cross-validation results for Scenario B. LightGBM ranks first by the predefined selection metric with PR-AUC = 0.4239. It also achieves the highest F1-score (0.4746) and ROC-AUC (0.7824). Random Forest is second in PR-AUC at 0.4218, followed by XGBoost at 0.4181 and Logistic Regression at 0.4170. XGBoost has the highest Accuracy at 0.8699, while Decision Tree has the lowest PR-AUC at 0.3838. Because model selection is based on mean cross-validation PR-AUC, LightGBM is passed to the tuning stage.

Table 5 reports test-set performance of the five default models. XGBoost has the highest test PR-AUC (0.4625), only 0.0015 ahead of LightGBM (0.4610), whereas LightGBM has the highest F1 (0.5131), Recall (0.5690), and Balanced Accuracy (0.7433). These test-set differences are reported descriptively only and are not used for model selection. Under our protocol, model selection occurs earlier using mean five-fold CV PR-AUC; therefore LightGBM, which leads Scenario-B CV PR-AUC at 0.4239, is the model passed to RandomizedSearchCV. Random Forest (0.4470) and Logistic Regression (0.4454) follow on

Table 5. Test-set performance of the five default models in Scenario B (duration excluded), each at its F1-optimized decision threshold from the validation stage (column-wise best among the five default models in bold). The selection rule selects LightGBM by mean CV PR-AUC for subsequent hyperparameter tuning. The last row reports the tuned LightGBM selected by cross-validation and evaluated at the validation-selected threshold.

Model	Thresh.	Accuracy	Bal. Acc.	PR-AUC	F1	Recall	Prec.
XGBoost	0.3997	0.8803	0.7373	**0.4625**	0.5099	0.5528	0.4732
LightGBM	0.4785	0.8784	**0.7433**	0.4610	**0.5131**	**0.5690**	0.4673
Random Forest	0.5775	0.8719	0.7359	0.4470	0.4964	0.5603	0.4456
Logistic Regression	0.5839	0.8752	0.7142	0.4454	0.4776	0.5065	0.4519
Decision Tree	0.5833	**0.8882**	0.6844	0.4046	0.4592	0.4213	**0.5045**
LightGBM (tuned)	0.6170	0.8818	0.7325	0.4658	0.5071	0.5399	0.4781

test PR-AUC, while the Decision Tree is weakest (0.4046). Accuracy is a poor guide here: the Decision Tree has the highest Accuracy (0.8882) and Precision (0.5045), but also the lowest Recall (0.4213) and the lowest Balanced Accuracy (0.6844).

LightGBM is selected for the Scenario-B tuning stage because it has the highest mean five-fold cross-validation PR-AUC (0.4239). The selected model is passed to `RandomizedSearchCV` with 12 candidate configurations and four-fold cross-validation using average precision as the objective. Threshold selection is then performed on the separate validation set using the procedure described in Section 3.7. This sequence keeps model selection, hyperparameter tuning, threshold selection, and final test evaluation separated.

The final LightGBM model uses a validation-selected threshold of 0.617, which also satisfies the Recall ≥ 0.50 check. On the held-out test set it achieves Accuracy = 0.8818, Balanced Accuracy = 0.7325, Recall = 0.5399, Precision = 0.4781, F1-score = 0.5071, PR-AUC = 0.4658, and ROC-AUC = 0.8049. Its confusion matrix contains 6,763 true negatives, 547 false positives, 427 false negatives, and 501 true positives.

The test PR-AUC is slightly *higher* than the CV PR-AUC for every model: 0.0284 for Logistic Regression, 0.0208 for Decision Tree, 0.0252 for Random Forest, 0.0444 for XGBoost, and 0.0371 for LightGBM. There is no base-rate shift between the two settings, since both are scored at the natural ratio of about 11.3%. A plausible reason is that the final models are fitted on the full training set rather than on four fifths of it. With only 928 positive test cases we do not over-interpret the size of the difference. What matters is that CV and test results agree in level and in model ordering.

4.2 Comparison charts

Figure 2 shows that Scenario A yields higher CV F1 than Scenario B for every model (a reduction of approximately 0.13–0.16 when duration is excluded). In

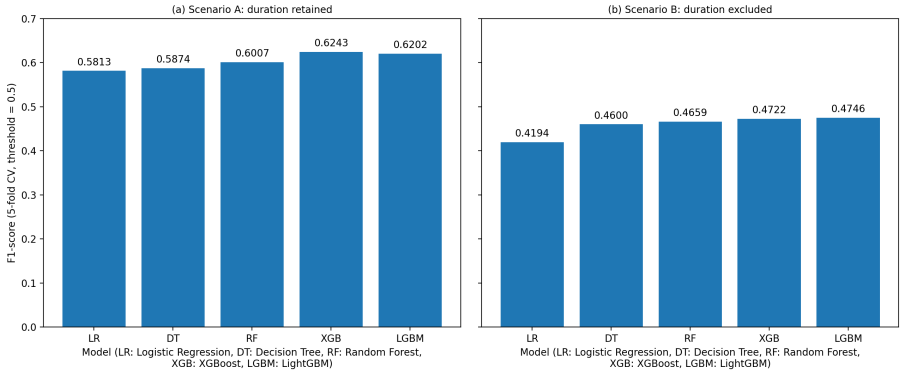


Fig. 2. F1-scores under stratified five-fold cross-validation on the training set: (a) Scenario A, duration retained; (b) Scenario B, duration excluded (SMOTE+Tomek inside each fold, default 0.5 threshold). LightGBM is the model selected by mean Scenario-B cross-validation PR-AUC.

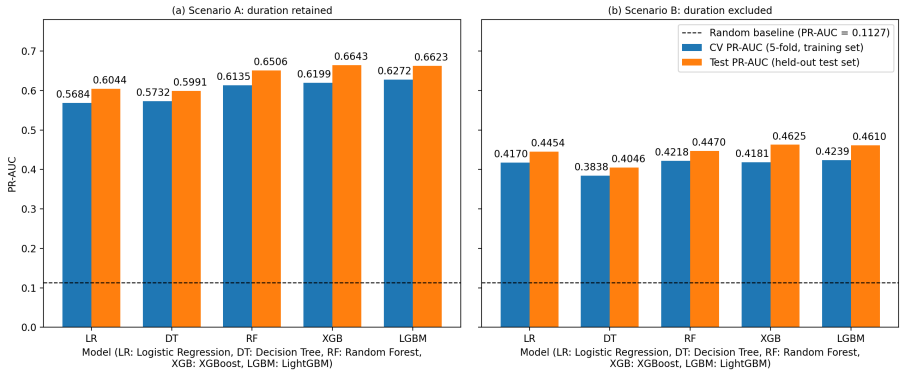


Fig. 3. PR-AUC of the five models in cross-validation versus the natural-ratio test set: (a) Scenario A, duration retained; (b) Scenario B, duration excluded. The dashed line marks the random baseline (PR-AUC = 0.1127).

Scenario B, LightGBM leads CV F1 (0.4746). Figure 3 shows that test PR-AUC exceeds CV PR-AUC by 0.0208–0.0444 across the five Scenario-B models; both sets retain the natural class ratio.

Figure 4 shows the model-specific precision–recall trade-off. The Decision Tree has the highest Precision (0.5045) but the lowest Recall (0.4213); Random Forest reaches Recall 0.5603 at Precision 0.4456. XGBoost and LightGBM both exceed 0.55 Recall while maintaining Precision above 0.46.

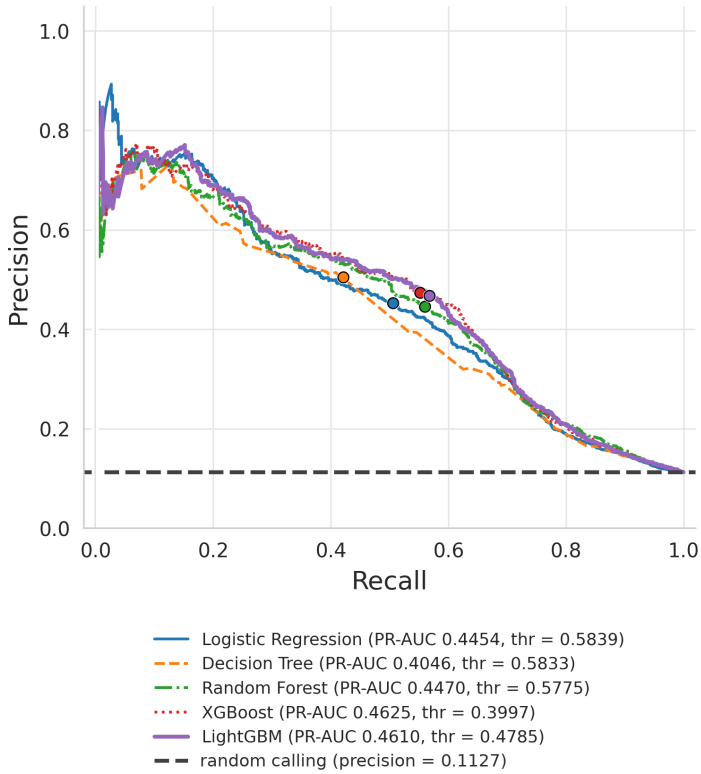


Fig. 4. Precision–Recall curves of the five default models on the Scenario-B test set. Circular markers indicate each model's operating point at its validation-selected F1-optimal threshold (Table 5). The dashed line marks the random-calling precision baseline of 0.1127.

4.3 Features with the strongest influence

The two-scenario analysis shows that call duration is highly informative after a call has taken place, but it is unsuitable for pre-contact prioritization. The final tuned LightGBM model in Scenario B assigns the largest gain share to `euribor3m` (74.63%), followed by `campaign` (8.78%), `pdays` (6.32%), `contact_cellular` (2.87%), and `season_q2` (1.76%). These values are model-specific gain importances rather than causal effects, and the concentration on `euribor3m` should therefore be interpreted as evidence that the final tree ensemble repeatedly uses this variable for predictive splits.

The final fitted pipeline selected the following twenty encoded features: `campaign`, `pdays`, `previous`, `euribor3m`, `has_prev_success`, `age_group_25_44`, `age_group_over65`, `job_blue-collar`, `job_student`, `job_unemployed`, `marital_married`, `marital_unknown`, `default_no`, `default_unknown`, `contact_cellular`, `contact_telephone`, `season_q1`, `season_q2`, `poutcome_nonexistent`, and `poutcome_success`.

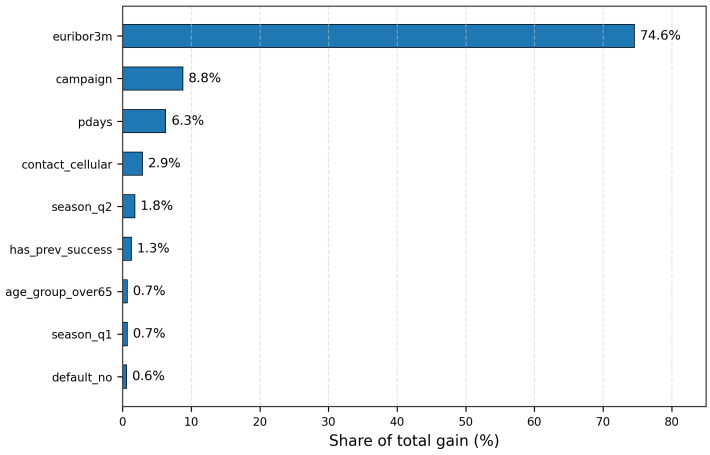


Fig. 5. Gain-based feature importance of the final tuned LightGBM model in Scenario B. The twenty features are selected by mutual information when the final pipeline is fitted on the complete training set; the plot shows the highest-ranked features among that final set.

Table 6. Ranked-calling performance of the final tuned LightGBM model on the Scenario-B held-out test set. Lift is calculated relative to the 11.27% positive-class base rate; calls avoided = 1 − calls/8,238.

Top (%)	Calls	Subscribers	Precision@k	Recall@k	Lift@k	Calls avoided
10	823	421	0.5115	0.4537	4.54	90.01%
20	1,647	607	0.3685	0.6541	3.27	80.01%
30	2,471	686	0.2776	0.7392	2.46	70.00%
50	4,119	761	0.1848	0.8200	1.64	50.00%

4.4 Effectiveness of ranked calling

The 8,238 held-out test customers were ranked by predicted subscription probability. Table 6 reports Precision@k, Recall@k, Lift@k, and calls avoided for four contact budgets. Calling the top 20% identifies 607 of 928 subscribers (Recall@20% = 0.6541; Lift = 3.27) while avoiding 80.01% of calls.

4.5 Comparison between Scenario A and Scenario B

Table 7 isolates the effect of removing call duration using the same default XGBoost configuration.

Removing duration reduces all four metrics, confirming its predictive value. Since duration is unavailable before contact, Scenario B remains the deployable setting; LightGBM is retained for tuning because it was selected by CV PR-AUC, not by test performance.

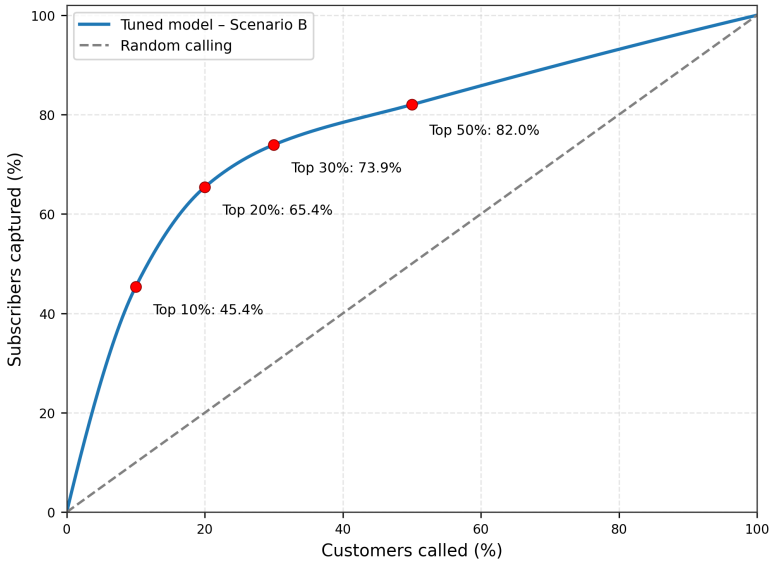


Fig. 6. Cumulative gain for ranked calling with the final tuned LightGBM model in Scenario B. The diagonal reference represents random calling at the natural class distribution.

Table 7. Like-for-like comparison of default XGBoost between Scenario A (duration retained) and Scenario B (duration excluded) on the held-out test set.

Metric	Scenario A (XGBoost)	Scenario B (XGBoost)	Difference (A − B)
PR-AUC	0.6643	0.4625	0.2018
F1	0.6600	0.5099	0.1501
Recall	0.7500	0.5528	0.1972
Precision	0.5893	0.4732	0.1161

5 Discussion

The common evaluation protocol makes the within-study model comparison more reliable than direct numerical comparisons with studies using different splits, feature sets, or imbalance treatments. LightGBM was selected using Scenario-B CV PR-AUC (0.4239), not test performance. Although default XGBoost has a marginally higher test PR-AUC than default LightGBM (0.4625 versus 0.4610), this descriptive difference does not justify changing the predefined selection rule. Tuning increases LightGBM PR-AUC to 0.4658; its F1 changes from 0.5131 to 0.5071 because the final model uses a validation-selected threshold.

The scenario comparison confirms that duration contains substantial predictive information but cannot support pre-contact targeting. In Scenario B, `euribor3m` accounts for 74.63% of gain, followed by `campaign` (8.78%) and `pdays` (6.32%). Gain importance describes model use rather than causal influence. The concentration on `euribor3m` may partly reflect campaign-period conditions, reinforcing the need for chronological validation.

The top-20% ranking captures 65.41% of observed subscribers, but this retrospective result is not a causal uplift estimate. The data come from one Portuguese bank in 2008–2010 [12]; random splitting may distribute campaign periods across subsets. Future work should use chronological holdout and live or randomized campaigns to test generalization and incremental impact. No formal significance test was conducted, so small differences between models should be interpreted descriptively.

6 Conclusion

This study compared Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM on the imbalanced UCI Bank Marketing dataset using a leakage-aware protocol. The two-scenario design distinguishes a post-contact reference from a deployable setting that excludes call duration.

In Scenario B, LightGBM achieved the highest mean five-fold CV PR-AUC (0.4239) and was selected for tuning. At the validation-selected threshold of 0.617, the final model achieved test PR-AUC 0.4658, F1 0.5071, Recall 0.5399, Precision 0.4781, Balanced Accuracy 0.7325, and ROC-AUC 0.8049. Ranking customers by its pre-contact scores captured 607 of 928 subscribers in the top 20% while avoiding 80.01% of calls.

The results support model-based prioritization but do not establish causal uplift. The single-bank, 2008–2010 dataset and random holdout limit generalization; chronological validation and a prospective campaign are needed to assess deployment impact.

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